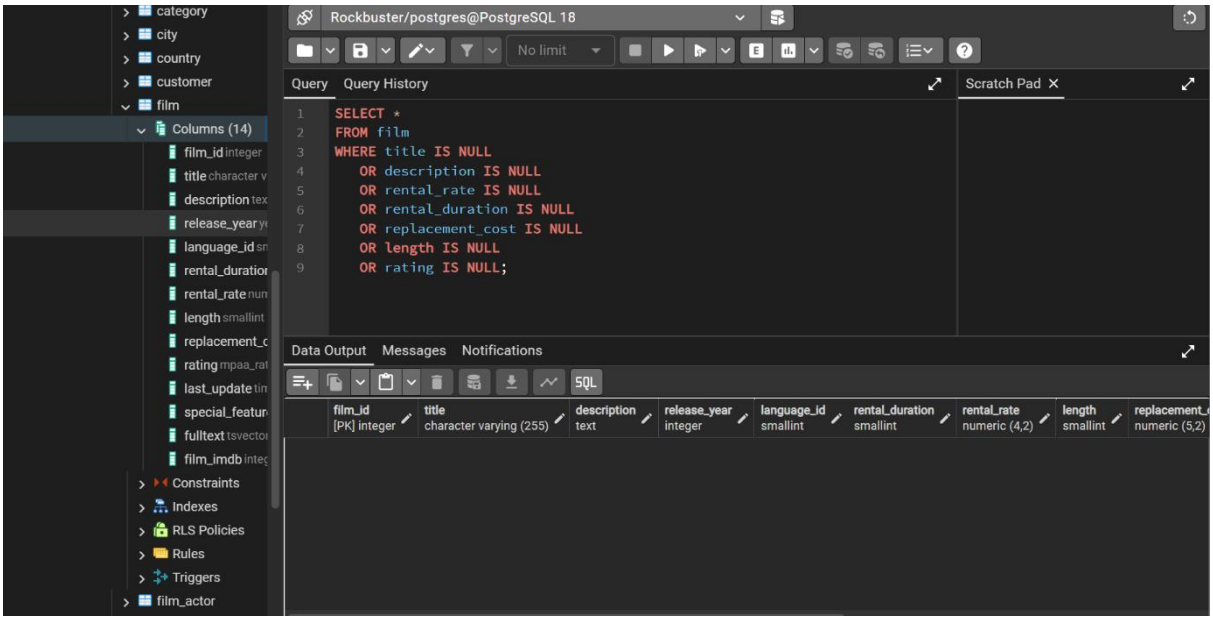
Summarizing & Cleaning Data in SQL

Check for and clean dirty data: Find out if the film table and the customer table contain any dirty data, specifically non-uniform or duplicate data, or missing values. Create a new “Answers 3.6” document and copy-paste your queries into it. Next to each query write 2 to 3 sentences explaining how you would clean the data (even if the data is not dirty).

Query applied:



Observation:

The query returned zero rows, which means there are no missing values in the columns checked (title, description, rental_rate, rental_duration, replacement_cost, length, rating). This indicates that the film table is well-structured and consistent, with no incomplete records.

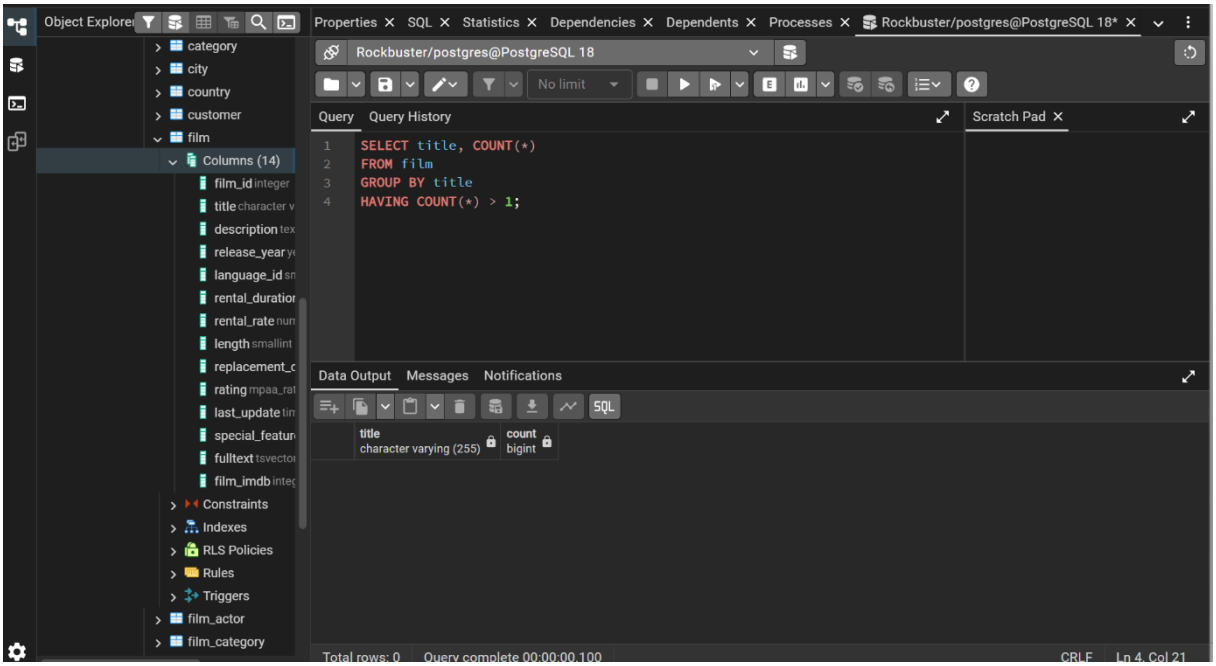
If NULL values had been present, I would clean the data by either:

Replacing missing values using logical defaults (for example, an average rental rate), or

Removing records only if the missing data made the row unusable for analysis.

Handling missing data is important because NULL values can distort calculations and lead to incorrect analytical results.

Query Applied:



Observation:

The query returned no results, meaning there are no duplicate film titles in the film table. This suggests that the dataset maintains good data integrity and avoids redundancy.

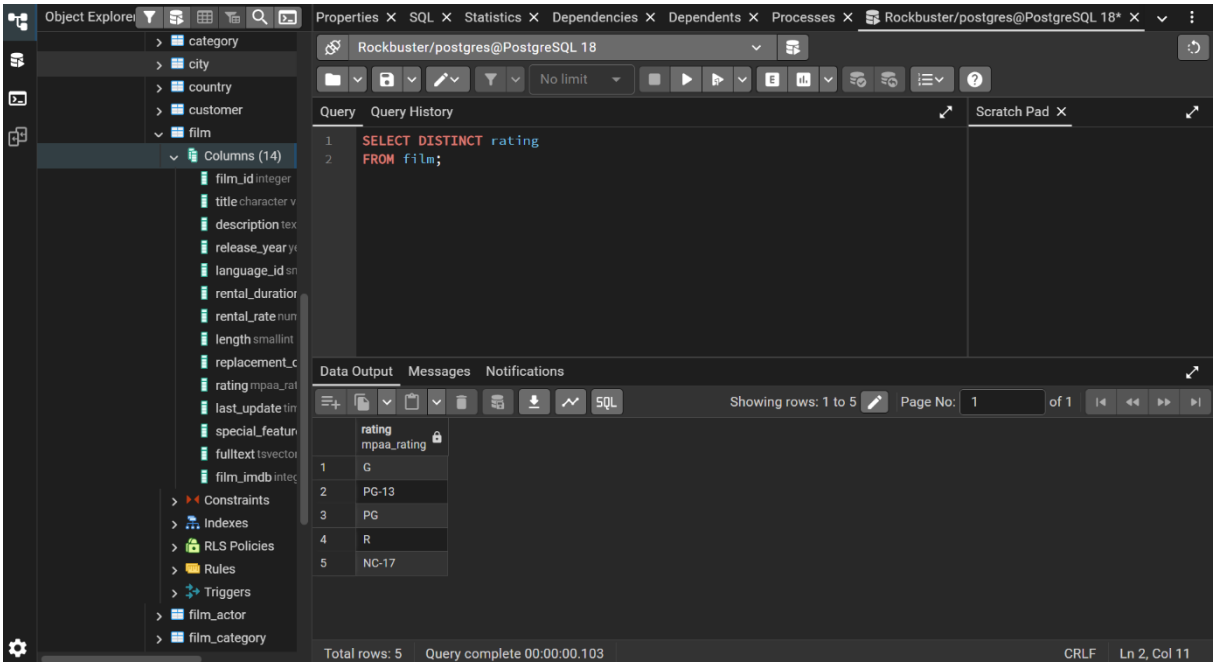
If duplicates had been found, I would clean the data by:

Investigating whether duplicates were true errors or legitimate entries,

Removing redundant rows while keeping the correct record.

Duplicate data is problematic because it can inflate counts, distort averages, and produce misleading business insights.

Query Applied:



Observation:

The query returned five distinct ratings: G. PG, PG-13, R and NC-17

This indicates that the rating column contains uniform and standardized values. No unexpected or inconsistent categories (such as misspellings or mixed formats) were found.

If non-uniform values had appeared, I would clean the data by:

Standardizing categories (for example, converting “Pg” → “PG”),

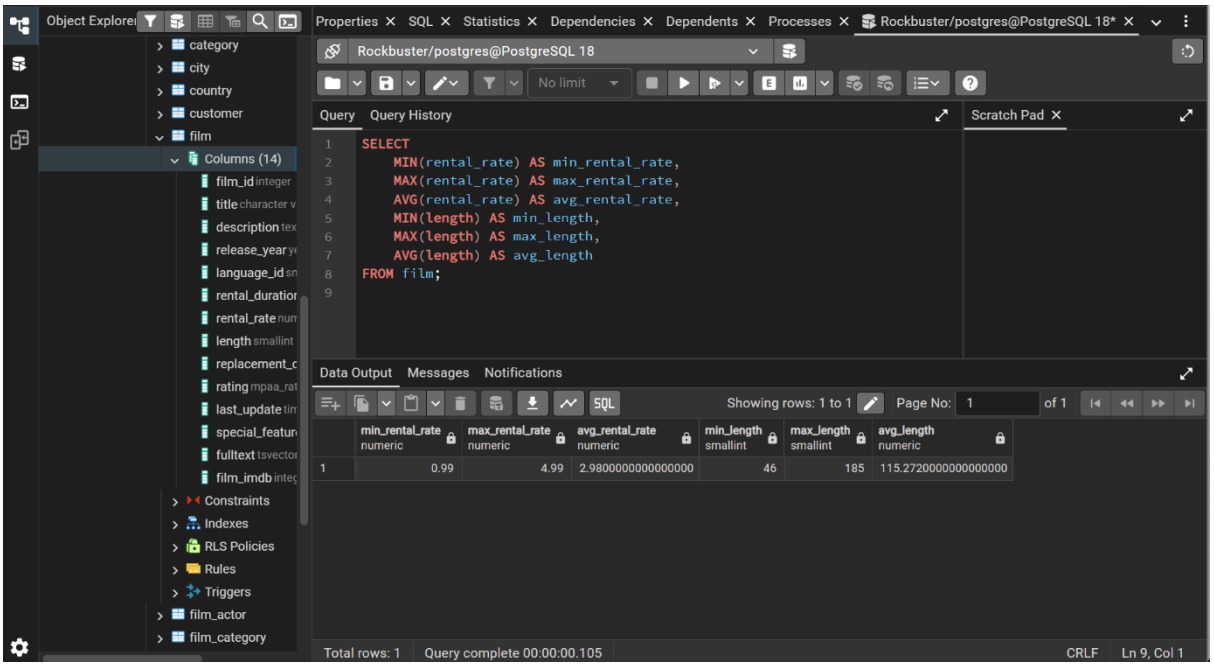
Correcting spelling errors or invalid entries.

Uniform categorical data is important because inconsistencies can break grouping operations and produce incorrect aggregations.

Overall Conclusion:

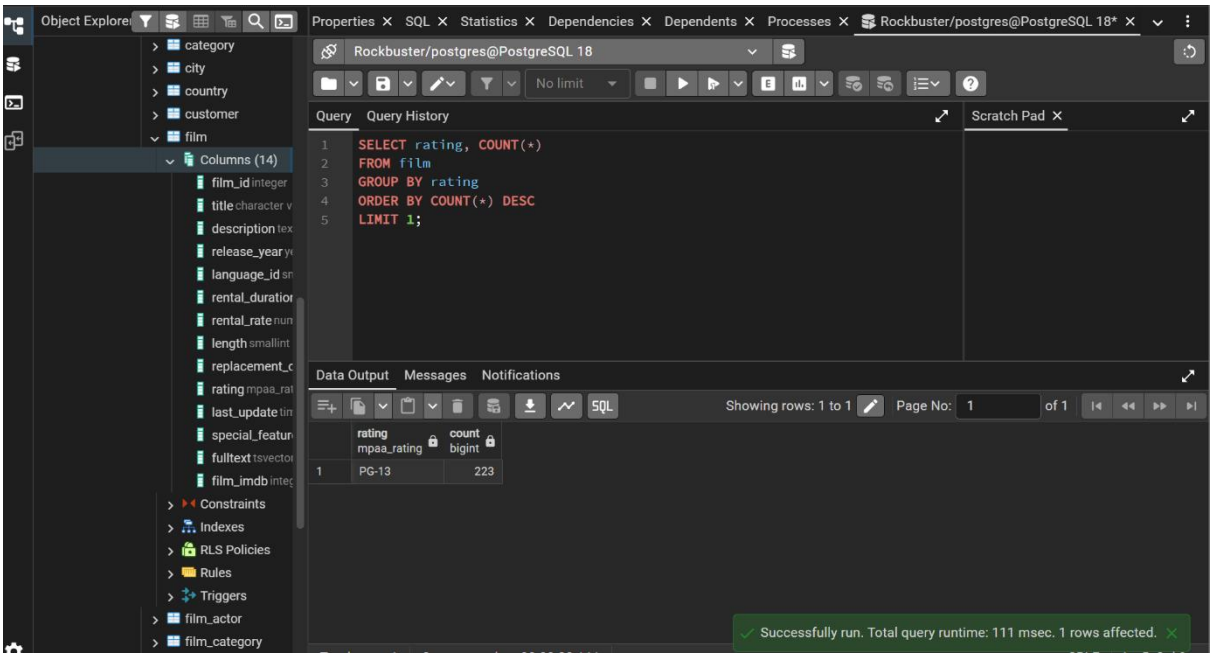
Based on the queries performed, the film table does not show signs of dirty data. There are no missing values, duplicates, or non-standard categories. This suggests the dataset is reliable and suitable for further analysis.

Summarize your data: Use SQL to calculate descriptive statistics for both the film table and the customer table. For numerical columns, this means finding the minimum, maximum, and average values. For non-numerical columns, calculate the mode value. Copy-paste your SQL queries and their outputs into your answers document.



Query: Film Table – Numerical

```
SELECT
    MIN(rental_rate) AS min_rental_rate,
    MAX(rental_rate) AS max_rental_rate,
    AVG(rental_rate) AS avg_rental_rate,
    MIN(length) AS min_length,
    MAX(length) AS max_length,
    AVG(length) AS avg_length
FROM film;
```



Query: Film Table – Mode

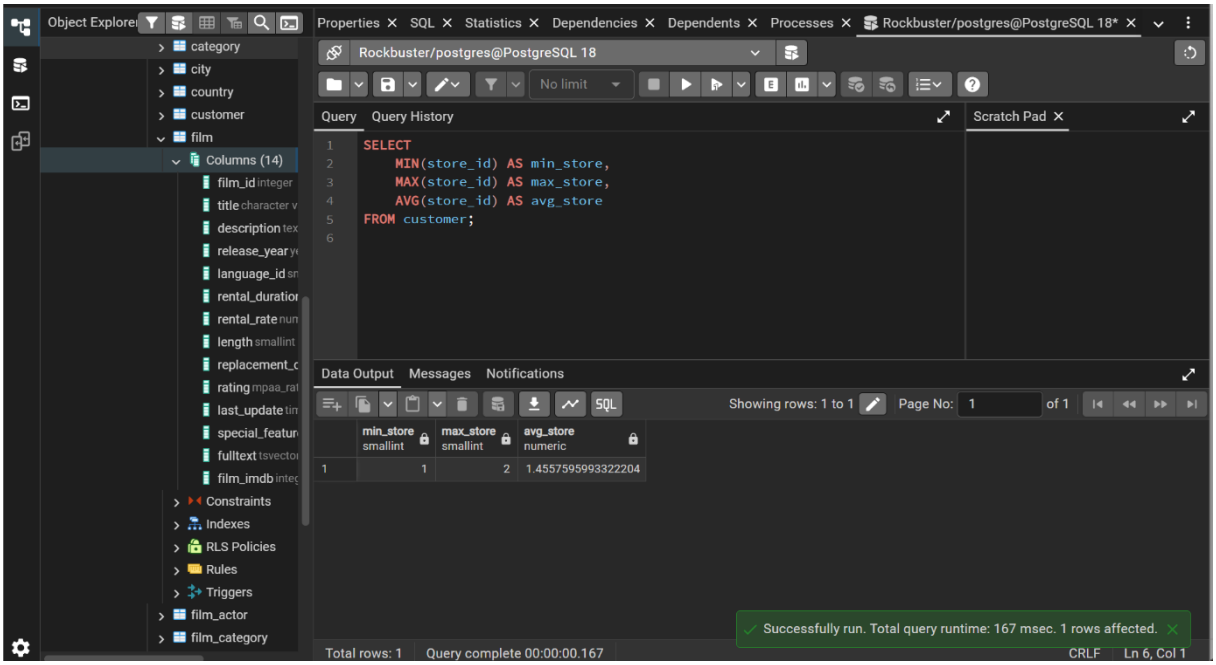
SELECT rating, COUNT(*)

FROM film

GROUP BY rating

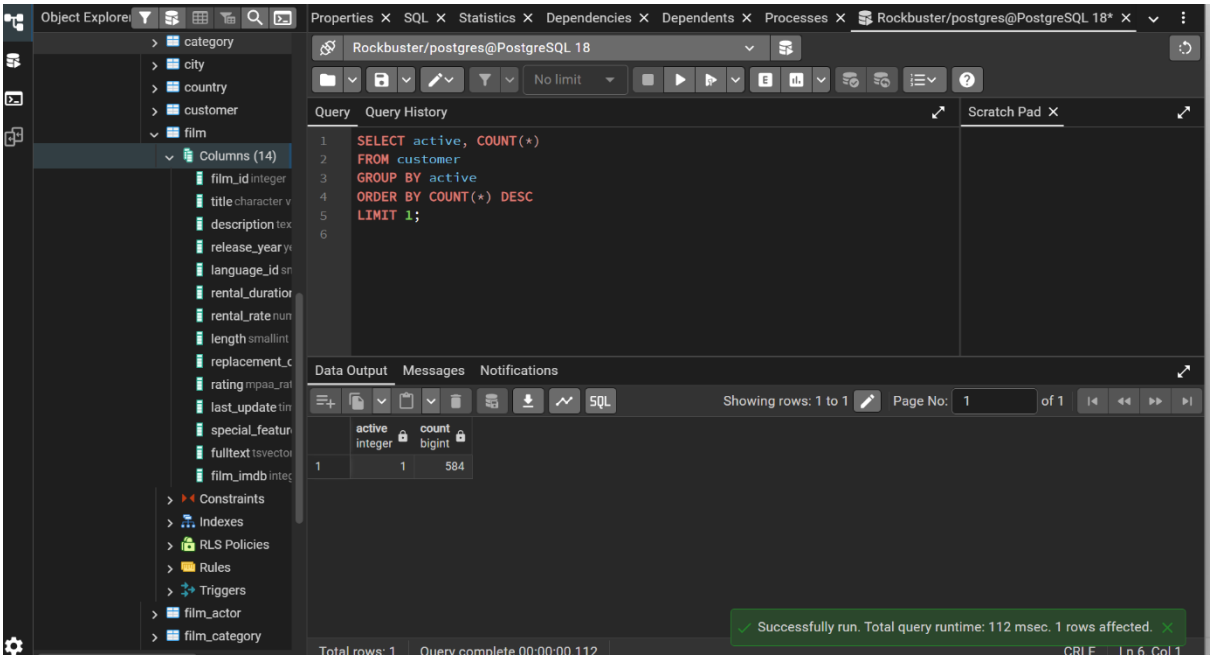
ORDER BY COUNT(*) DESC

LIMIT 1;



Query: Customer Table – Numerical

```
SELECT
   MIN(store_id) AS min_store,
   MAX(store_id) AS max_store,
   AVG(store_id) AS avg_store
FROM customer;
```



Query: Film Table – Mode

```
SELECT active, COUNT(*)
FROM customer
GROUP BY active
ORDER BY COUNT(*) DESC
LIMIT 1;
```

Reflect on your work: Back in Achievement 1 you learned about data profiling in Excel. Based on your previous experience, which tool (Excel or SQL) do you think is more effective for data profiling, and why? Consider their respective functions, ease of use, and speed. Write a short paragraph in the running document that you have started.

SQL is generally more effective for data profiling when working with large datasets because it is faster, scalable, and designed for structured querying. It allows analysts to compute statistics, detect anomalies, and filter records efficiently without performance degradation. Excel is more visual and user friendly for smaller datasets, but it becomes slower and harder to manage as data volume increases. SQL also reduces manual errors by automating checks through queries rather than relying on spreadsheet formulas.